

Hassanat from Mu'tah University in Jordan is the creator of this technology.

The software reads the movements of lips to authenticate access to the user. The efficiency of the system is as high as 80 per cent. The user has to speak out the words to get verified. The system contains two stages. The user can set a visual password and verify himself or herself. The system is equipped with a VSR system that recognises words and extracts the sequence of feature vectors. When the user tries to log-in, these features are digitally compared with the visual password, after which the system grants access to him or her.

Hassanat tested the system with video database from over 20 people. The error rate in the workability of the system is just 7.63 to 20.51 per cent. Hassanat claims that this technology has the potential to become the most secure security system as even actors cannot mimic the lip movements of other people.

Nano-structure technology that replicates human brain

A unique nano-structure is being created using a thin film of oxide material by a team of researchers led by Dr Sriram from RMIT University. The material that they are using to develop this nano-structure is 10,000 times thinner than human hair. Researchers are further working on the development of hard drive technology using this nano-structure. This could be the new leap in the field of memory storage devices.

New devices made using this nano-structure can be faster as well as reliable. The thin film is capable of creating a memristive effect. The device can log the history of events using the memristive effect. This effect can be used to train the computer to behave like human brain. It is very similar to synaptic interfaces in the human brain.

Researchers are testing the structure with various electronic devices. The current structure has a data-storage size of 60 nanometres. They are also experimenting with implementation response time of the human neural network.

Innovative bionic ear that can switch on or off

AT&S, a leader in advanced packaging solutions, Soundchip SA, a Swiss-based innovator in wearable sound technology and STMicroelectronics, a global semiconductor leader serving customers across the spectrum of electronics applications, have collaborated for crafting a bionic hearing module that, when installed into a personal audio device, delivers an amazing wearable sound experience controlled at the ear by the wearer and software intelligence.

Personal audio devices, like an MP3 player or smartphone, equipped with the bionic hearing module, provide wearers with the ability to electronically 'open' and 'close' their ears to ambient sound conditions, or even to augment ambient sound with programmed audio from a connected smart device. This capability can fully protect wearers from noise in situations where the ambient sound is too loud, or to open the ear for natural conversation with others, without having to remove the audio device, suffer from the discomfort of occlusion, or worse, the pain of loud noise.

The bionic hearing module integrates a broad spectrum of advanced electronics to further enhance the on-the-go audio experience, including head-tracking and other sensing, to enable exciting new features, including augmented-audio guidance and biometric monitoring.

The multi-mode audio capabilities of the bionic hearing module are enabled through the use of HD-PA technology developed by Soundchip. Their implementation in a compact form factor is made possible through the use of patented Soundstrate technology, which enables the efficient combination of electronic, acoustic and transmission means within a single, compact mechanical structure.

The semiconductor components in the bionic hearing module comprise the latest motion and audio MEMS (micro electro mechanical system) components from STMicroelectronics, an HD-PA-compliant audio engine for zero-latency

sound processing, and an ultra-low-power STM32 MCU from ST's industry-leading portfolio of more than 500 32bit ARM Cortex-M-core microcontrollers.

The bionic hearing module's packaging employs the latest in embedded component packaging and 2.5D PCB technology from AT&S, which is capable of integrating acoustic, electro-acoustic, passive and active electronic components with unmatched efficiency, providing module dimensions ideally suited to the comfort and size constraints of in-ear operation, and is compatible with most existing in-ear-type personal audio devices.

Wearable skin to monitor the heart and skin

A new device has been developed by scientists that is just like our skin. When put on, it works as a fitness-monitoring device, by keeping a check on our heart and skin. Researchers from Northwestern University and University of Illinois described the tool as a medical device which is enabled to raise an alert whenever any cardiovascular trouble occurs or skin gets dry and requires moisturising.

This wearable, wireless skin-like device can function 24 hours a day. The device is 5cm, which has around 3600 liquid crystals on a thin, soft and flexible substrate that can be placed directly on the skin. According to Yonggang Huang, who led the theory, design and modelling aspect of the research, the device is 'mechanically invisible.' He also called it first-of-its-kind.

Transient temperature change happens at the skin's surface, which is used by the technology in the device. This temperature change determines blood-flow rate which is connected to cardiovascular health and skin hydration levels. Huang also said that whenever temperature is felt by a crystal, its colour changes. Its dense array provides a snapshot of the distribution of temperature across that particular area of the device. The temperature data is changed into an accurate health report by an algorithm within 30 seconds. A wireless heating system is also a part of the device which gets powered by electromagnetic waves in the air and it determines thermal properties of the skin.

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